

Study Plan to Paper 1 Publication

Edmond Mwinbamon Balika | BCEPS / IGS, University of Bergen | Target: Paper 1 Submission - months 12–15

1. Phased Study Plan (Months 1–15)

Phase	Learning Focus	Resources	Study Actions & Deliverable	UiB Foundational Coursework
Months 1–3 Epidemiology & Disease burden	How incidence, prevalence, mortality; DALY / YLL / YLD are derived at sub-national level as per GBD logic; descriptive vs causal claims	A: GBD 2023 Diseases & Injuries; Causes of Death; Risk Factors A: WHO Global Health Estimates A: Intro epidemiology text/course	- Health Metrics Notebook (defs, equations, Ghana examples) - Worked DALY examples - Deliverable: can explain and interpret GBD outputs	- Complete personal-data e-course (Mitt UiB) - Enrol MEDMET901 Ethics & health research (7 ECTS; aim Year 1) - Join CIH–CISMAC / IGS research school - Confirm elective plan with supervisors (≤30 ECTS total)
Months 2–5 GIS fundamentals	Vector/raster, CRS, joins, clipping, reprojection; Ghana admin units; proximity vs travel time vs effective access	B: QGIS Training Manual B: A Gentle Introduction to GIS E: Ghana 2021 PHC; DHS 2022; GHS/MoH facility docs	- QGIS drills on Ghana boundaries, WorldPop, roads, facilities - Clean region/district map project - Deliverable: reproducible Ghana spatial project (QGIS)	- MEDMET901 sessions / home exam as scheduled - Research-school seminars (global health / methods) - Begin logging dissemination hours toward FORMIDL901
Months 4–7 R spatial & reproducibility	sf, terra, ggplot2, exactextractr; zonal stats; spatial allocation; Git + project structure	B: R spatial docs (sf, terra, ggplot2, exactextractr) E: WorldPop; census; admin layers; DHIS2 (if accessible)	- Scripted Ghana spatial database - Join population ↔ admin units - Version-controlled paper1/ pipeline - Deliverable: sf/terra workflow joining population & admin data	- Keep coursework bounded: protect 15–18 h/week research time - Align any methods electives with GIS/epi gaps (supervisor OK)
Months 5–9 Access, inequality, uncertainty	Friction surfaces; dry/rain travel time; concentration index, SII/RII, Theil; uncertainty in the pipeline	B: Spatial accessibility literature (travel time / access) E: OSM roads; friction/seasonal layers; DHS stratifiers A: revisit GBD uncertainty concepts as needed	- Pilot accessibility metric (1 region/districts) - One inequality measure + short interpretation note - Uncertainty hooks in code (not end add-on) - Deliverable: working Paper 1 pilot + reproducible pipeline	- Research-school presentations / internal talks (dissemination credits) - Mid-way evaluation planning only if timing falls in window
Months 9–15 Paper 1 production	Freeze core RQ & analysis plan; scale pilot nationally; validation; package method; draft & submit	A + B + E as needed for methods/results write-up Writing: applied spatial-burden papers as models Defer deep C & D (costing / CEA) until Paper 2–3	- National core analysis + validation + uncertainty - R package / reproducible method - Manuscript draft early; iterate with supervisors - Deliverable: Paper 1 submitted (~ mo. 12–15)	- Use Paper 1 seminar/conference abstract toward dissemination ECTS - Do not stack new heavy electives during write-up

2. Resource Priority for Paper 1

Block	Priority to Paper 1	Usage
A. Epidemiology & disease burden	Essential	Core reading path for Months 1–3 and GBD interpretation throughout Paper 1
B. GIS & spatial health	Essential	QGIS → R spatial → accessibility literature in sequence above
E. FairChoices & Ghana evidence	Essential (Ghana data)	Census, DHS, GHS/MoH, DHIS2 for spatial layers & validation; FairChoices depth can wait for Paper 3
C. Health economics & costing	Defer	Light awareness only; start after Paper 1 submission
D. Equity, CEA	Defer	Conceptual skim optional; deep study in Year 2–3 for Paper 3

3. Progress Checkpoints

By	Must be true	Evidence
Month 3	Explain epidemiologic/burden measures disaggregation; interpret GBD outputs	Health Metrics Notebook + short oral/written check with supervisor
Month 6	Acquire, clean, join and map principal Ghana spatial datasets	Scripted maps + README in gis/ / data/
Month 9	Working Paper 1 pilot + reproducible analysis pipeline	Pilot repo run that regenerates key figures
Month 12–15	Paper 1 submitted	Submission confirmation + deposited preprint/supplement if used

4. Weekly Blocks During UiB Course Weeks

Activity	Normal Week	Heavy Course Week (e.g. MEDMET901)
Research / Data Analysis	15–18 h	8–10 h
Reading (A/B/E)	5 h	3 h (course reading counts toward A where overlapping)
GIS / Spatial Practice	4 h	2 h
UiB Courses / BCEPS Research	2–4 h	15–20 h (block weeks)
Academic writing	4 h	2 h
Supervision	1–2 h	1 h